

07_Exploritory

Contents

When loading in the data I have determined from the events table, the variables Sex, NOC, Season, and medals are all factor variables and needed to be coded this way in R.

Table 1

Table 2 The next set of two by twos is for Summer and Winter Medal by Sports

```
summer1 <- MedalWinners("Summer", 1896)
summer2 <- MedalWinners("Summer", 2016)

list(summer1,summer2) %>%
  kableExtra::kable("latex", position = "H", booktabs = T,
    label = "SummerWinners",
    caption = "Number of medal winners in 1896 versus 2016 Summer Olympic Games") %>%
  kable_styling(latex_options = "striped")
```

Table 1: Number of medal winners in 1896 versus 2016 Summer Olympic Games

Sport	Medals_Given	Sport	Medals_Given
Athletics	37	Archery	24
Cycling	16	Athletics	192
Fencing	9	Badminton	24
Gymnastics	37	Basketball	72
Shooting	15	Beach Volleyball	12
Swimming	10	Boxing	51
Tennis	10	Canoeing	82
Weightlifting	6	Cycling	84
Wrestling	3	Diving	36
		Equestrianism	45
		Fencing	65
		Football	106
		Golf	6
		Gymnastics	66
		Handball	89
		Hockey	99
		Judo	56
		Modern Pentathlon	6
		Rhythmic Gymnastics	18
		Rowing	144
		Rugby Sevens	74
		Sailing	45
		Shooting	45
		Swimming	191
		Synchronized Swimming	32
		Table Tennis	24
		Taekwondo	32
		Tennis	24
		Trampolining	6
		Triathlon	6
		Volleyball	72
		Water Polo	78
		Weightlifting	45
		Wrestling	72

Table 2: Number of medal winners in 1924 versus 2014 Winter Olympic Games

Sport	Medals_Given	Sport	Medals_Given
Alpinism	21	Alpine Skiing	31
Bobsleigh	13	Biathlon	60
Cross Country Skiing	6	Bobsleigh	24
Curling	16	Cross Country Skiing	60
Figure Skating	12	Curling	24
Ice Hockey	28	Figure Skating	45
Military Ski Patrol	12	Freestyle Skiing	30
Nordic Combined	3	Ice Hockey	130
Ski Jumping	3	Luge	24
Speed Skating	16	Nordic Combined	18
		Short Track Speed Skating	43
		Skeleton	6
		Ski Jumping	21
		Snowboarding	30
		Speed Skating	51

```
winter1 <- MedalWinners("Winter", 1924)
winter2 <- MedalWinners("Winter", 2014)

list(winter1,winter2) %>%
  knitr::kable("latex", position = "H", booktabs = T, caption = "Number of medal winners in 1924 versus 2014 Winter Olympic Games",
    label = "WinterWinners") %>%
  kable_styling(latex_options = "striped")
```

Summer Breakdown

The next two by twos is for Summer and Winter Medal Winners by Teams

```
summer1 <- MedalBreakDownByTeam("Summer", 1896)

summer1 %>%
  kableExtra::kable("latex", position = "H", booktabs = T,
    label = "SummerBreakDownWinners1",
    caption = "Medal allocation for 1896 Summer Olympic Games",) %>%
  kable_styling(latex_options = "striped", font_size = 7)
```

Table 3: Medal allocation for 1896 Summer Olympic Games

Team	None	Gold	Silver	Bronze	TotalEvents	TotalMedals
Greece	96	10	16	18	140	44
Germany	62	24	5	2	93	31
United States	7	11	7	2	27	20
France	15	5	4	2	26	11
Great Britain	16	2	3	2	23	7
Denmark	9	1	2	3	15	6
Hungary	12	2	1	3	18	6
Austria	3	2	1	2	8	5
Switzerland	5	1	2	0	8	3
Australia	2	2	0	0	4	2
Australia/Great Britain	0	0	0	2	2	2
Ethnikos Gymnastikos Syllogos	0	0	0	2	2	2
Great Britain/Germany	0	2	0	0	2	2
Greece-1	0	0	2	0	2	2
Greece-2	2	0	0	0	2	0
Greece-3	2	0	0	0	2	0
Italy	1	0	0	0	1	0
Sweden	5	0	0	0	5	0

```
summer2 <- head(MedalBreakDownByTeam("Summer", 2016), 30)

summer2 %>%
  kableExtra::kable("latex", position = "H", booktabs = T,
    label = "SummerBreakDownWinners2",
    caption = "Medal allocation for 2016 Summer Olympic Games for the top 30 teams",) %>%
  kable_styling(latex_options = "striped", font_size = 7)
```

Table 4: Medal allocation for 2016 Summer Olympic Games for the top 30 teams

Team	None	Gold	Silver	Bronze	TotalEvents	TotalMedals
United States	443	137	52	67	699	256
Germany	371	47	43	67	528	157
Great Britain	325	64	55	26	470	145
Russia	285	50	28	35	398	113
China	374	44	30	35	483	109
France	408	20	55	21	504	96
Australia	428	23	34	25	510	82
Italy	325	8	38	24	395	70
Canada	328	4	4	61	397	69
Japan	372	17	13	34	436	64
Serbia	73	14	27	13	127	54
Brazil	525	34	6	6	571	46
Netherlands	276	9	25	11	321	45
Spain	310	7	19	17	353	43
Denmark	102	15	10	16	143	41
New Zealand	196	6	25	5	232	36
Jamaica	47	11	17	2	77	30
Sweden	158	2	23	3	186	28
Croatia	69	7	15	2	93	24
South Korea	231	13	3	8	255	24
South Africa	132	2	7	14	155	23
Argentina	206	21	1	0	228	22
Hungary	182	12	3	7	204	22
Belgium	117	2	17	2	138	21
Norway	58	0	0	19	77	19
Azerbaijan	51	1	7	10	69	18
Kazakhstan	105	3	5	10	123	18
Nigeria	59	0	0	18	77	18
Poland	269	3	3	10	285	16
Romania	101	4	2	10	117	16

```
winter1 <- MedalBreakDownByTeam("Winter", 1924)

winter1 %>%
  kableExtra::kable("latex", position = "H", booktabs = T,
    label = "WinterBreakDownWinners1",
    caption = "Medal allocation for 1924 Winter Olympic Games",) %>%
  kable_styling(latex_options = "striped", font_size = 7)
```

Table 5: Medal allocation for 1924 Winter Olympic Games

Team	None	Gold	Silver	Bronze	TotalEvents	TotalMedals
Great Britain	11	16	0	11	38	27
Norway	19	4	7	6	36	17
Finland	18	4	8	3	33	15
United States	29	1	10	1	41	12
Canada	8	9	0	0	17	9
Sweden	32	1	8	0	41	9
France	48	0	0	8	56	8
India	0	7	0	0	7	7
Belgium-1	0	0	0	5	5	5
Switzerland	26	4	0	1	31	5
Austria	0	3	1	0	4	4
Great Britain-1	2	0	4	0	6	4
Switzerland-1	0	4	0	0	4	4
France-1	4	0	0	2	6	2
Australia	0	1	0	0	1	1
Nepal	0	1	0	0	1	1
Belgium	27	0	0	0	27	0
Czechoslovakia	31	0	0	0	31	0
France-2	6	0	0	0	6	0
Great Britain-2	6	0	0	0	6	0
Hungary	6	0	0	0	6	0
Italy	14	0	0	0	14	0
Italy-1	5	0	0	0	5	0
Italy-2	5	0	0	0	5	0
Latvia	7	0	0	0	7	0
Poland	15	0	0	0	15	0
Switzerland-2	4	0	0	0	4	0
Yugoslavia	7	0	0	0	7	0

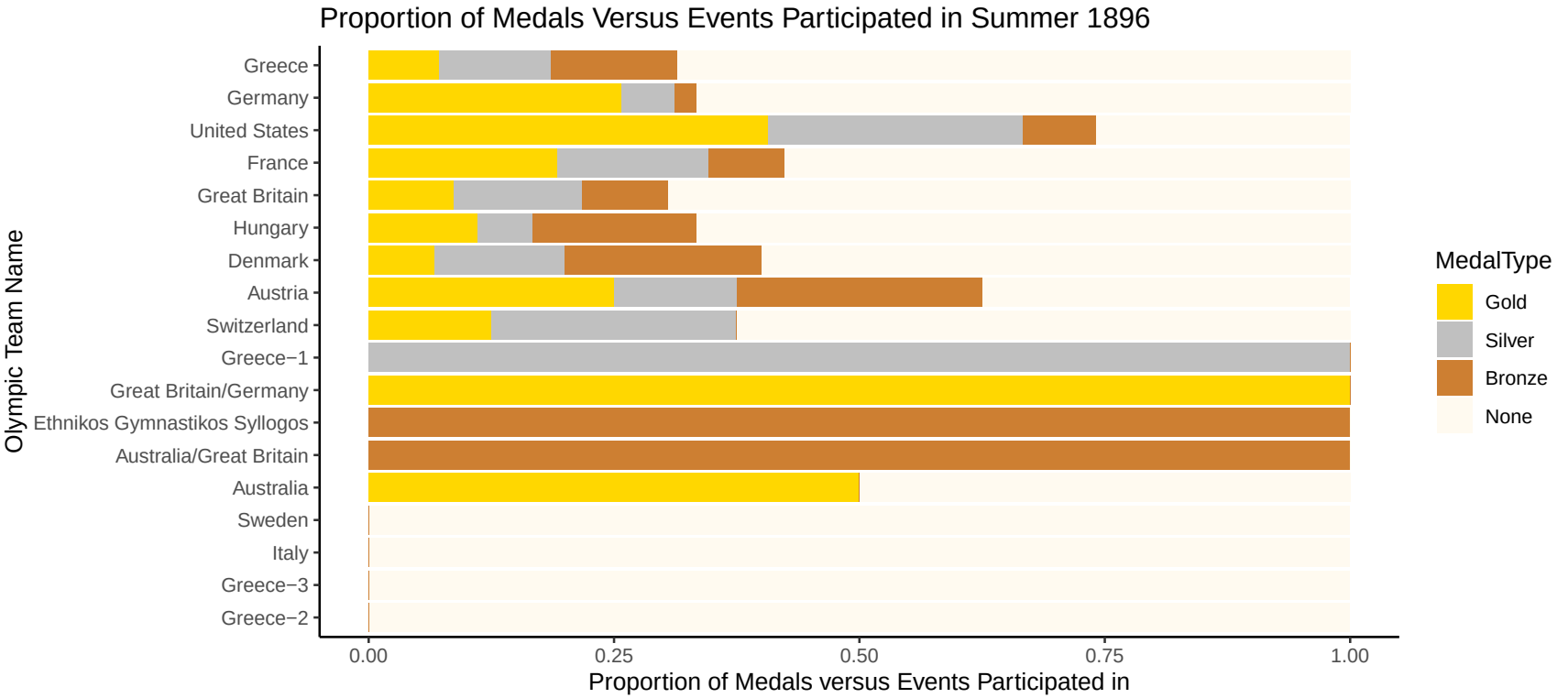
```
winter2 <- head(MedalBreakDownByTeam("Winter", 2014),30)

winter2 %>%
  kableExtra::kable("latex", position = "H", booktabs = T,
    label = "WinterBreakDownWinners2",
    caption = "Medal allocation for 2014 Winter Olympic Games for the top 30 teams",) %>%
  kable_styling(latex_options = "striped", font_size = 7)
```

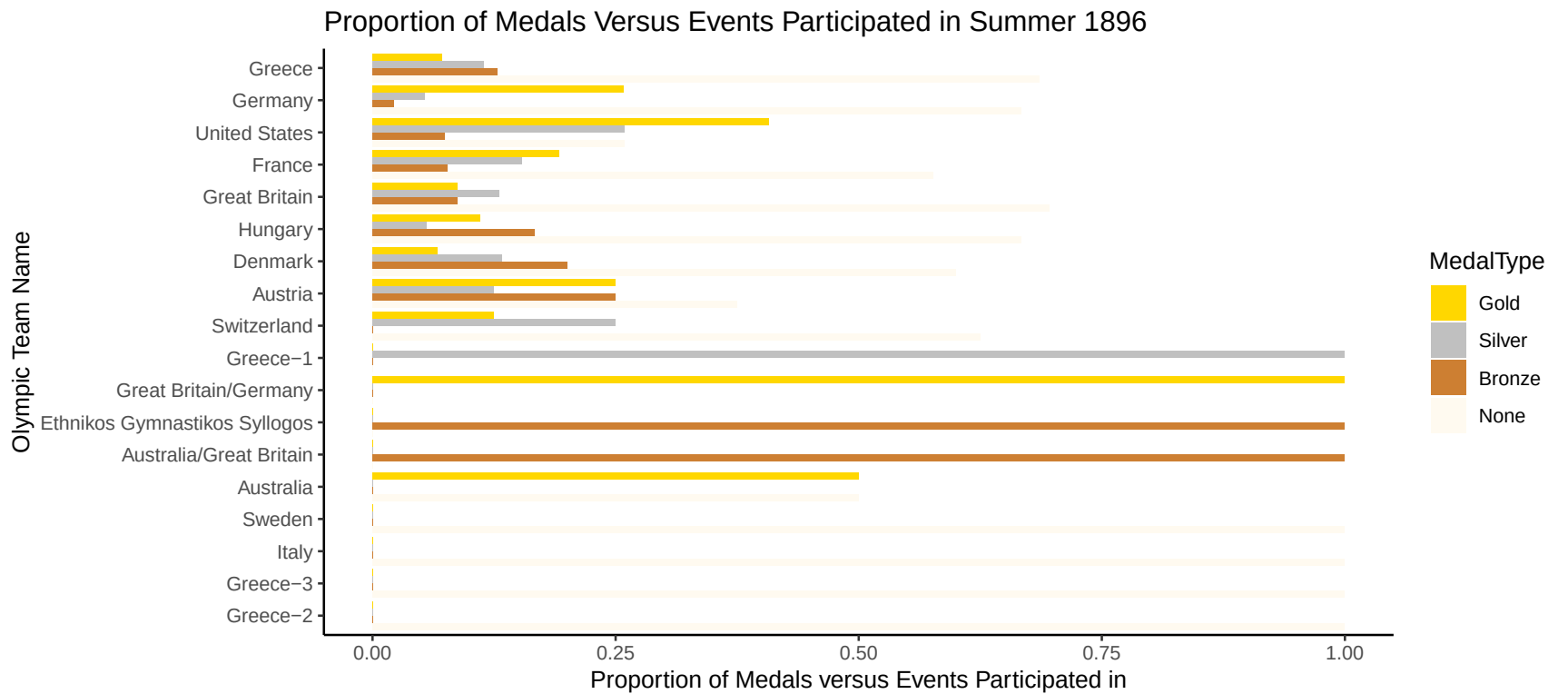
Table 6: Medal allocation for 2014 Winter Olympic Games for the top 30 teams

Team	None	Gold	Silver	Bronze	TotalEvents	TotalMedals
Canada	241	57	20	5	323	82
Russia	261	25	20	11	317	56
United States	281	8	28	16	333	52
Sweden	97	8	32	11	148	51
Norway	183	18	5	13	219	36
Finland	114	2	7	24	147	33
Germany	203	13	12	7	235	32
Netherlands	53	13	7	9	82	29
Switzerland	187	6	2	20	215	28
Austria	184	4	10	11	209	25
France	171	4	4	10	189	18
Italy	201	0	2	12	215	14
South Korea	104	7	5	2	118	14
China	93	3	4	5	105	12
Czech Republic	168	2	7	2	179	11
Japan	160	1	4	6	171	11
Poland	138	4	4	3	149	11
Great Britain	60	1	4	5	70	10
Russia-1	4	8	0	2	14	10
United States-1	4	2	2	6	14	10
Slovenia	108	2	2	4	116	8
Belarus	66	5	0	1	72	6
Latvia-1	2	0	4	2	8	6
Latvia	61	0	1	4	66	5
Ukraine	96	4	0	1	101	5
Canada-1	8	2	2	0	12	4
Germany-1	10	2	0	2	14	4
Australia	81	0	2	1	84	3
Austria-1	0	0	2	0	2	2
Russia-2	12	0	2	0	14	2

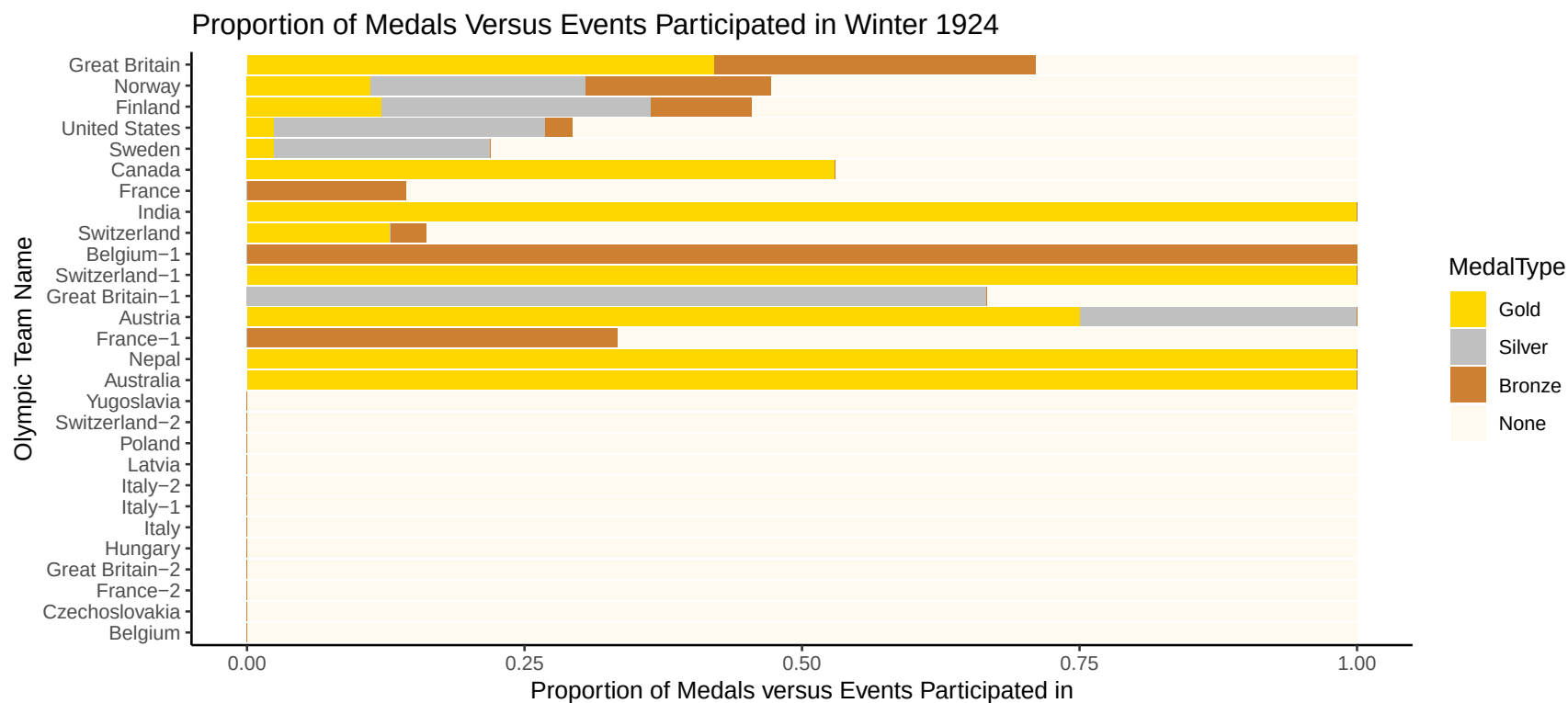
PropTeamWinners("Summer", 1896, "Fill", 30)



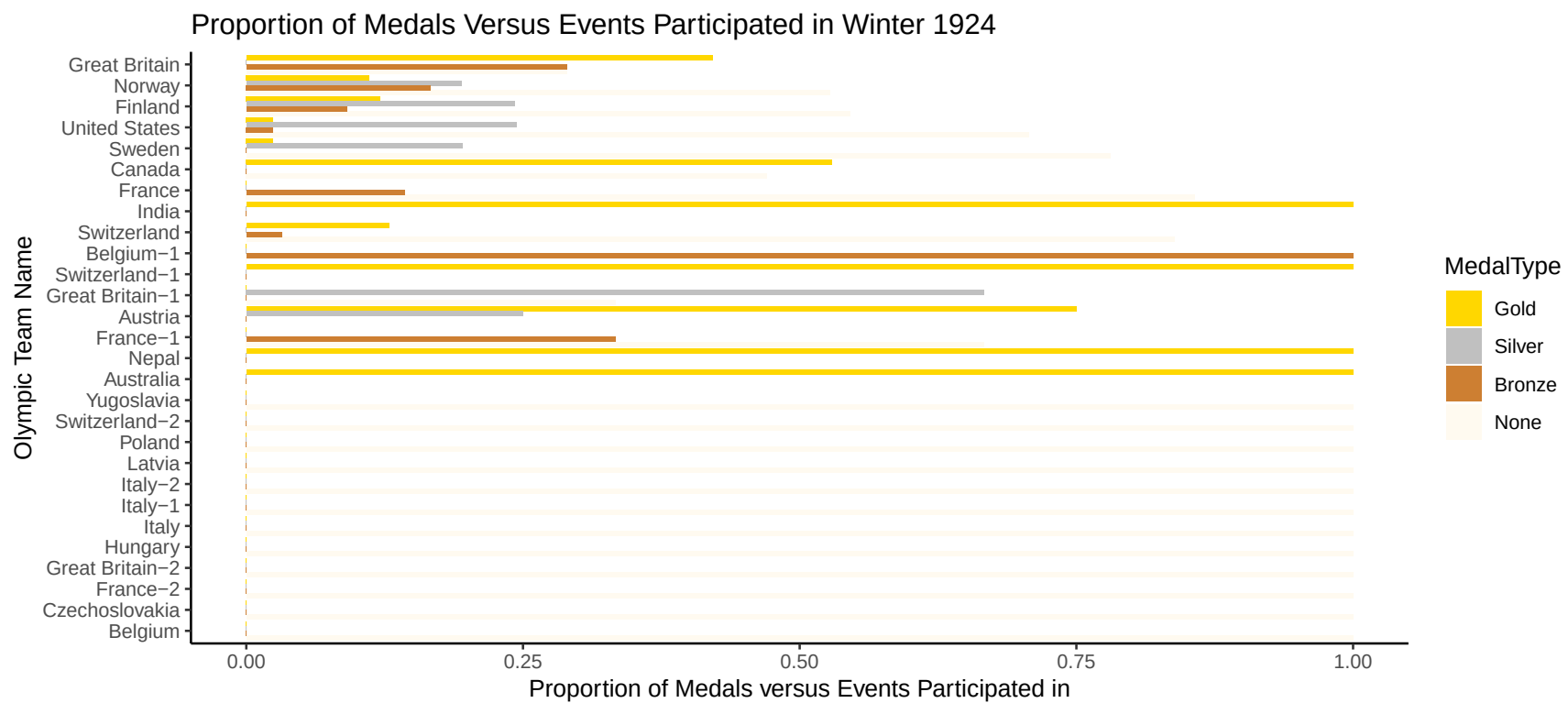
PropTeamWinners("Summer", 1896, "Dodge", 30)



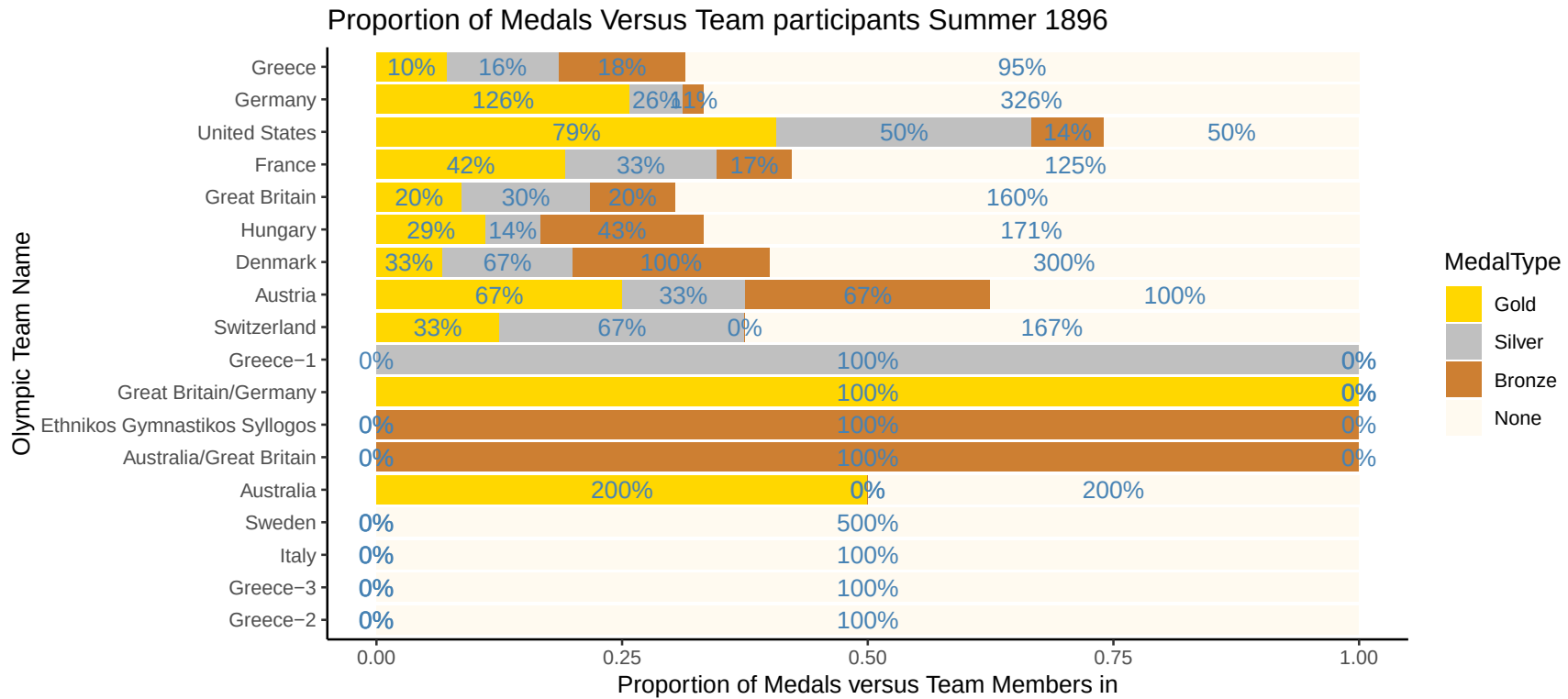
```
PropTeamWinners("Winter", 1924, "Fill", 30)
```



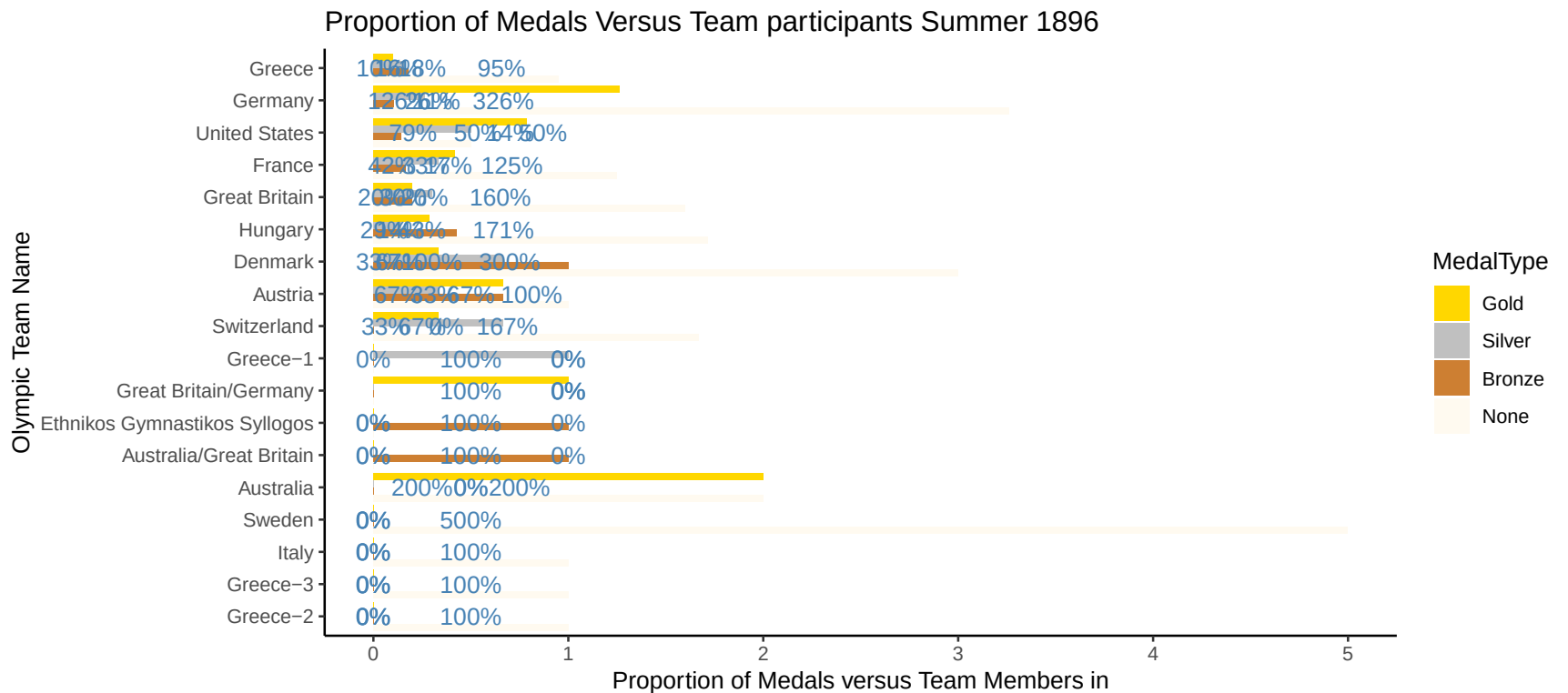
```
PropTeamWinners("Winter", 1924, "Dodge", 30)
```



```
PropTeamAthleteWinners("Summer", 1896, "Fill", 30)
```

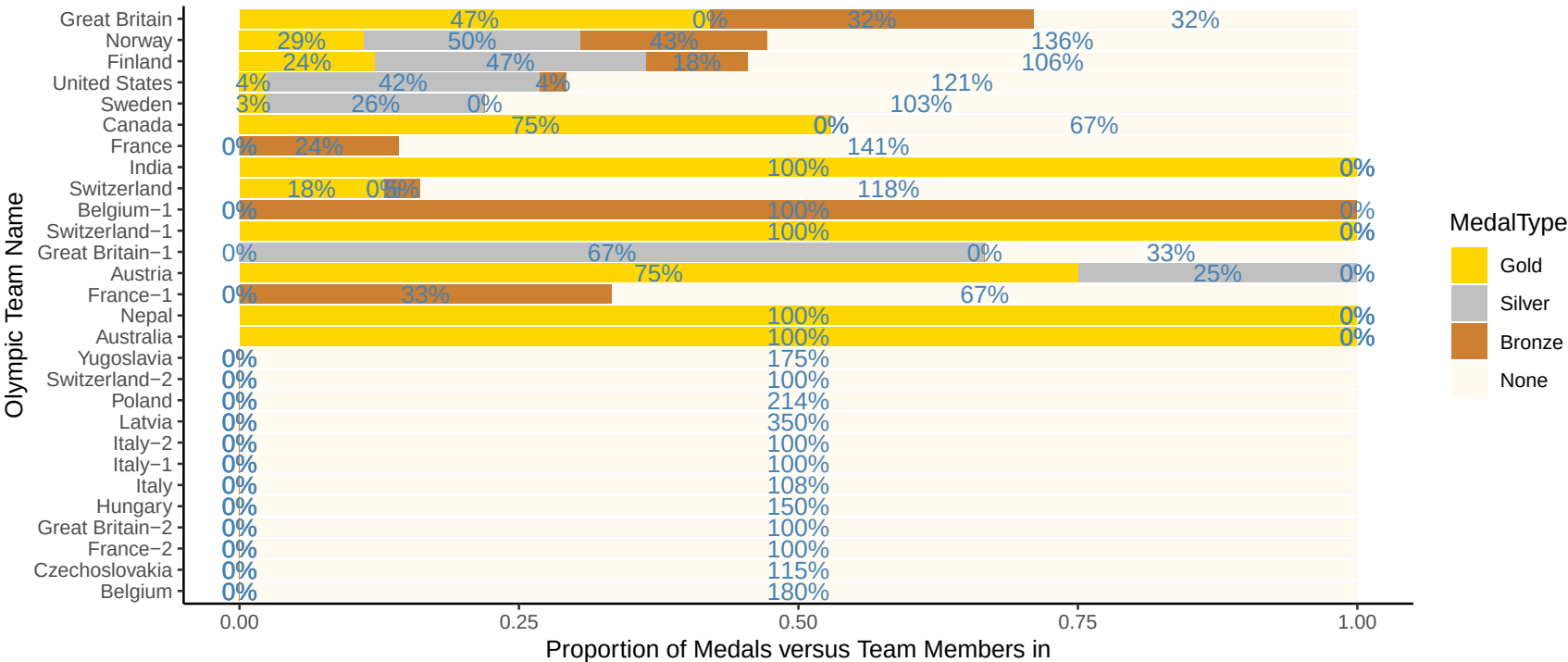


```
PropTeamAthleteWinners("Summer", 1896, "Dodge", 30)
```



```
PropTeamAthleteWinners("Winter", 1924, "Fill", 30)
```

Proportion of Medals Versus Team participants Winter 1924



PropTeamAthleteWinners("Winter", 1924, "Dodge", 30)

Proportion of Medals Versus Team participants Winter 1924

